

Organs, Not Weights

Label-Free Capability Transplantation Across a 15,000× Model Gap

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Abstract. We introduce *organ grafting*: label-free transplantation of computational capability from a 27-billion-parameter teacher into a living 1.713-million-parameter $O(1)$ -state host whose identity must remain unchanged. The transplant is a 5,732–14,625-parameter structured *organ* that wraps the frozen host as a logit-space shell, reads embeddings and scan state, and writes only onto the candidate slice of the output distribution. No host weight, recurrence, tokenizer, or hidden dimension is shared with the donor. Organs are harvested from candidate-restricted teacher distributions, sharpened at $\tau = 2$, and trained in seconds on CPU with zero ground-truth labels. Mounting and unmounting are exact; the host remains bit-identical at language NLL 8.6656 throughout.

A single 14,625-parameter dual organ performs held-out addition and subtraction at 1.000 accuracy across seeds. A 5,732-parameter recurrent accumulator trained only on two-operator chains scores 1.000 on unseen lengths 3, 4, 6, 8, and 12, while a matched flat control collapses at length 3. The same structural program climbs further: a shift-structured stack exposes capacity as an exact slot-count step function and expands from four to eight slots with zero retraining; two $O(1)$ counters decide $a^n b^n c^n$ with two learned parameters; and a two-parameter multiplicative fingerprint recognizes the copy language ww , with a sharp float32 capacity edge at position 25. These organs exceed their donor wherever structure supplies the algorithm.

The governing result is the *Quotient Law*: a free head generalizes exactly onto the symmetry quotient of its training lattice. On relation tasks, held-out accuracy equals the orbit coverage parameter-free and cell-by-cell; on selection tasks, the orbit quantity remains a lower bound and is broken precisely when the choice criterion factorizes class-locally. True composition therefore comes from structural form, not free-head interpolation. *Crystallization* completes the transfer by extracting the learned invariant and replacing its approximate readout with the exact realization over the task’s group: integer line, log lattice, or finite circle. It raises donor-distilled chains to 1.000 through length 12, extends arithmetic beyond the donor’s answer space, and turns invariant fit into a label-free authenticity and group-classification gate. The final cold OrganBank routes four crystallized organs through a 384-parameter state router and exact parser, answers 152 of 152 measured tasks plus 30 of 30 text windows, generates all 96 multi-digit products sequence-exact, and survives host int4 deployment unchanged. This establishes label-free, reversible capability transplantation across a 15,000× model gap into a frozen constant-memory host.

Keywords. capability transplantation; label-free distillation; modular sequence models; state-space models; structural generalization; crystallization; $O(1)$ state.

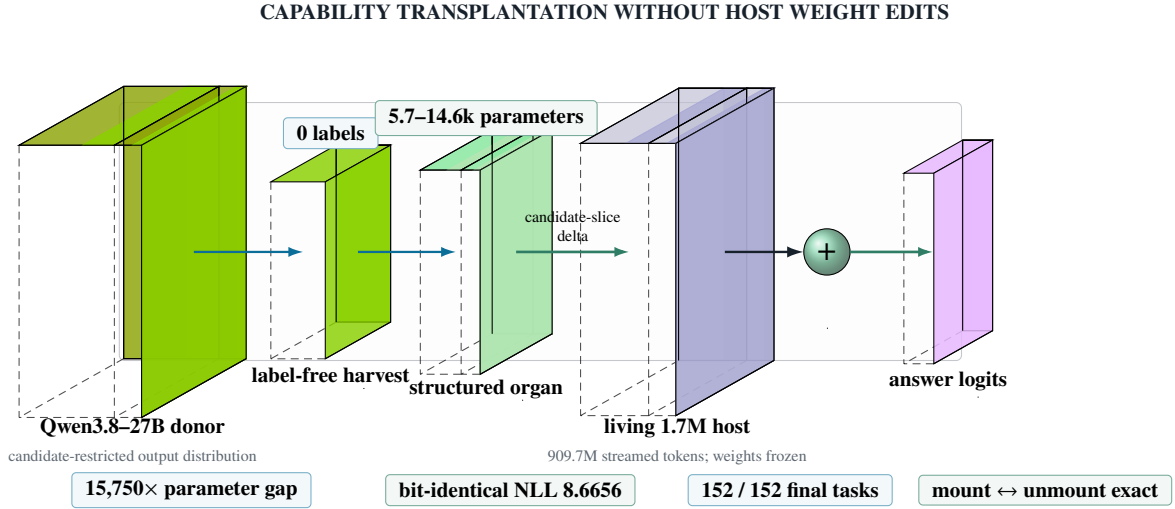


Figure 1. Organ grafting in one pass. A 27B donor supplies only a candidate-restricted output distribution. A small structured organ distills the capability, wraps a frozen 1.7M O(1)-state host as a logit-space shell, and writes only onto the task’s answer slice. The host remains bit-identical; the final banked system answers all 152 measured tasks correctly.

1 The Setting

1.1 The host is alive, and it stays frozen

The recipient is not a randomly initialized network and not a fine-tuning substrate. It is **o1-state** (Foss, 2026a,b): an O(1)-state sequence organism built on the GSSM selective-scan operator, trained by streaming — checkpoint A1, **1,713,673 parameters**, having lived through **909,676,544 tokens** of C4 at constant memory. Its identity is a measured quantity: WikiText-2 validation NLL **8.6656**. Every experiment in this paper preserves that identity exactly. The host’s weights are frozen; mounting an organ never edits them; and unmounting is verified **bit-identical** in every run. In the final system the host is not even mounted — the organ operates as a mount-free shell around it, and the host forward pass is untouched by construction.

This is the transplantation problem in its hardest form: the recipient has a life, a language distribution, and theoretical guarantees (the GSSM contraction bound, Theorem 2 of (Foss, 2026a)) that must all survive the operation.

1.2 The donor is four orders of magnitude larger

The donor is **Qwen3.8-27B** (4-bit GGUF, llama.cpp), a model **$\sim 15,750\times$ larger** than the host. We take from it *no weights, no activations, no labels* — only its output distribution over a small candidate set of answer words, collected on 352 prompts per task. Everything the organ learns, it learns from that distribution.

1.3 Why no one else does this

Every existing transfer axis is a different problem:

- **Knowledge distillation** shrinks a teacher into a student trained *from scratch or from a compatible init* — the student’s identity is the distillation. Our host’s identity predates the transfer and must survive it untouched.
- **Quantization-aware distillation (QAD)** (Liquid AI, 2026) moves a model along the *precision* axis (BF16 $\rightarrow$ int4) under the assumption that the capability already exists in the student. Our axis is *capability*: the host scores at chance on the task before the graft.
- **Task arithmetic and model merging** operate between models of the *same architecture and size* by editing weights. Our donor and host share no architecture, no tokenizer, no dimension — and the host’s weights are never edited.
- **Adapters and LoRA** fine-tune with ground-truth labels inside the host’s forward pass. We use zero labels, and the host forward pass is never modified.

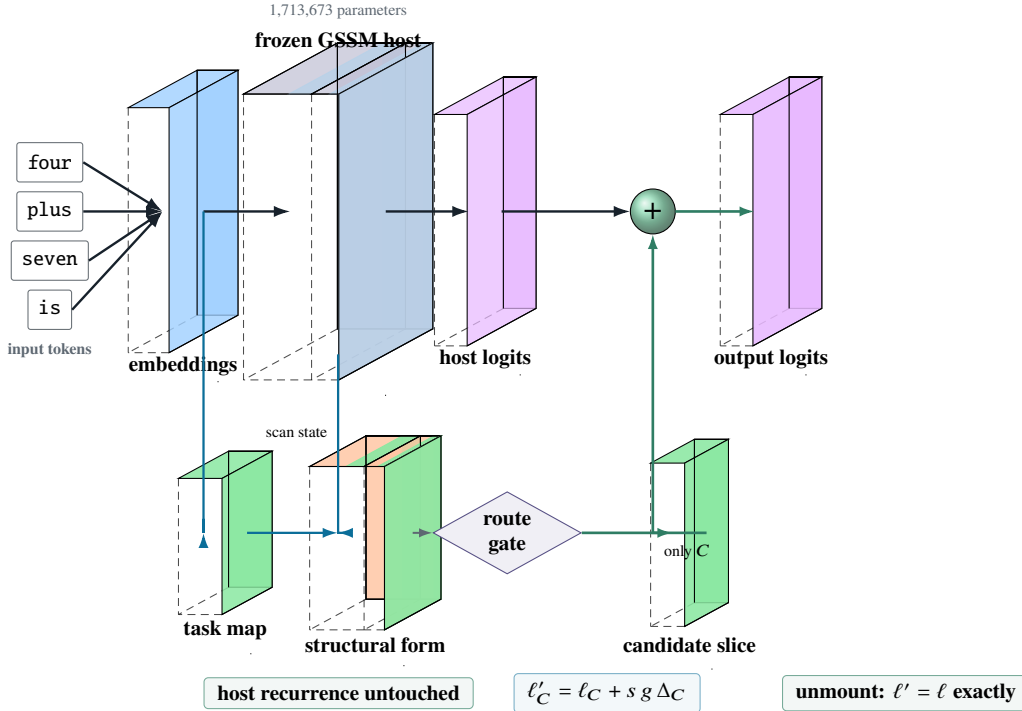


Figure 2. OrganShell architecture. The living host produces its ordinary logits and state under frozen weights. The organ reads embeddings and scan state, maps them into the task algebra, applies the structural form and a jurisdiction-only route gate, and adds a delta solely to the candidate slice C . No host layer or recurrence is replaced.

The axis measured here — placebo-controlled, label-free capability transplantation across four orders of magnitude into a frozen, living, $O(1)$ -state host — is served by no prior system.

2 Method

2.1 OrganShell: capability lives in logit space

An organ is a small structured module (5,732–14,625 parameters in this paper) wrapped *around* the frozen host as a shell. It reads the host’s token embeddings and scan state, and it writes a delta **only onto the candidate slice** of the output logits — the handful of answer words the task can produce. No host module is replaced; the GSSM contraction guarantee is untouched because the organ never enters the recurrence.

The shell placement is a measured result, not a preference. We first built organs into the token-velocity path of the scan (the route that preserves Theorem 2 by construction) and measured two structural negatives: a velocity organ confined to layer 0 cannot learn at all (the frozen head does not translate its pushes), and a velocity organ spanning both layers memorizes through the contextual layer-1 input (66k parameters: train 0.744, held-out 0.100). **Any route that sees the operand pair as a whole memorizes; the route that cannot see the pair cannot get through. Logit space with a structural constraint is the way out** (Sec. 4.1).

Language stays silent through a 4-gram gate regularized to zero on text (gate activation 0.0003); the host’s language NLL is bit-identical at 8.6656 in every headline run.

2.2 The Quotient Law

The central scientific object of this paper is a law that predicts, cell by cell and parameter-free, which held-out tasks an organ answers correctly. It was earned through two registered falsifications of its own predecessors (Sec. 4.3); we state it here in final form.

Setup. Training lattice $G \subset \mathbb{Z}^2$, answer $f(a, b) = a + \sigma b$, prediction by argmax over the candidate slice of the frozen host’s logits. Three head families:

- H_{sep} (free separable): $\delta_c = R_1(Pe_a) + R_2(Pe_b)$ — trivial symmetry group.

- H_{sym} (free symmetric): $\delta_c = r_c^\top P(e_a + e_b)$ — the AdditivOrgan; symmetry group $\mathbb{Z}_2$, $(a,b) \sim (b,a)$.
- H_{ord} (ordinally structured): $\delta_c = s(cu - qc^2)$ with $u = \alpha(a) + \sigma\beta(b)$ and *no free per-class parameters* — the hard form; the encoder form enforces the same structure through a 1D bottleneck.

Proposition 1 — free families memorize. On every lattice in this regime (up to 44 cells), the free families fit even *fully permuted* training targets: a real organ fits 23 permuted cells at 0.957 — identically at embedding width 32 and width 8 — and the bare head family does the same without host or gate. The mechanism is not linear interpolation (that hypothesis was registered and falsified): argmax correctness is an *inequality system* of $n(|C| - 1)$ margins, and the free families carry enough degrees of freedom (~ 170 against 23 cells) to solve it even under the sum constraint. Consequence: train = 1.000 carries zero information about held-out, the reading of additivity as “memorization impossible” is quantitatively refuted, and every generalization claim for a free head must be tested on truly unseen orbits.

Proposition 2 — the Quotient Law. A trained free head generalizes *exactly onto the S-quotient of its training lattice*:

$$z \text{ is correct} \iff \text{Orb}_S(z) \cap G_{\text{train}} \neq \emptyset. \quad (1)$$

The $\Leftarrow$ direction is exact: if (b,a) is in training, then (a,b) is *the same input* to a symmetric head — correctness is input identification, not generalization. The $\Rightarrow$ direction holds empirically with zero exceptions across 11 configurations: truly unseen cells always fall — on relation tasks; Proposition 4 states the exact scope. A synthetic sweep over square-lattice subsets makes the law quantitative: held-out accuracy equals the mirror share of the held set *exactly* — 1/5, 2/5, 9/10, 3/5 across four splits — deterministically over seeds, with host and gate demonstrably irrelevant: the phenomenon is intrinsic to head family plus SGD.

Proposition 3 — the error signature and the drift polytope. Cross-entropy on true labels pulls memorized solutions toward the ordinal attractor, so errors on unseen cells are ± 1 **on the answer axis or an escape to the candidate boundary** (the boundary class has only one-sided margins). Within H_{ord} the drift polytope is the *exact* misfit theory: writing the drift as $\varepsilon_a + \eta_b$, train-perfection is the linear system $|\varepsilon_a + \eta_b| \leq \frac{1}{2}$ over training cells; the vulnerability of a held-out cell is a linear program — dually a parity-weighted graph distance — with $V \geq 3/2$ for every unseen cell; and under the parabolic neighbor margin $k^2/2$ the cheapest excursion is exactly ± 1 . Whether a drift misfit is *realized* is decided by data force alone: balanced ground-truth data drives the drift to zero and yields held-out 1.000 despite $V = 3/2$.

Proposition 4 — the selection/relation dichotomy (law v4.1). The orbit criterion of Proposition 2 is a guaranteed *lower* bound — the $\Leftarrow$ direction is exact input identification and can never be lost. Whether the bound is *sharp* is decided by the task class. **Relation tasks** — the answer is a new class computed from both operands (add, sub) — make it sharp: the compositional solution requires global coordination of all class weights (the number line), and SGD finds memorization instead, so truly unseen orbits always fall. **Selection tasks** — the answer *is* one of the operands (max/min via *larger/smaller*) — break the bound structurally: the exact solution is locally realizable per class and closed-form constructible inside the free family — pair membership $r_c \cdot (x_a + x_b) = [c \in \{a,b\}]$ plus an operator tilt $\sigma_{\text{op}} \cdot \varepsilon \cdot c$, each class learning its own embedding matching independently — and that is precisely the gradient path cross-entropy prefers, so free heads generalize beyond the orbit. Round 5 puts the dichotomy on three task families per side (Sec. 3.2): add, sub, and mod-3 as relations; max, min, and median as selections. And the stack organ (Sec. 3.7) fixes the law’s scope with final precision: selection character makes the *task* free-learnable, never the *algorithm* — wherever a flat shortcut exists, SGD takes it, and only the structural prior buys the true procedure. Two further clauses complete the law (v4.2 and v4.3, both measured in Sec. 3.2). *The gate channel*: an organ’s free family is never bare — it is $\delta \times \text{gate} + \text{host}$, and the non-separable 4-gram gate computes: it rescues exactly the 3-class answer space (mod-3: held-out 0.857 against 0.000 with the gate off) and is dead from 4 classes upward — a step, not a slope (U-rescue 4/4 at $m=3$, 0/4 at $m=4, 5$, and 7, the orbit bound exact at 13; Sec. 3.2, Round 7) — so the granularity axis is binary, and a strict organ needs a gate that only routes. *Factorizability*: selection splits into membership, which always generalizes, and choice among members, which generalizes freely exactly when the criterion factorizes class-locally — pair-coupled criteria behave relationally even though the answer is an operand. The design consequence

survives every extension sharpened: **only the structural form that carries the task’s group composes class-independently — the parabola over $\mathbb{R}$ for ordinal tasks, the circle for cyclic ones, the log map for multiplicative ones.**

Consequences.

1. **No anchor set repairs a free head.** Neither the diagonal nor a full anchor row $a=9$ rescues the triangular lattice (both measured: held-out 0.000 in both width regimes). “More data” is the wrong axis; the only thing that saves a cell is its own orbit — and on a triangle the mirror of every cell lies outside the lattice. That is why the triangle collapses, and why the square merely *looked* solved.
2. **Apparent composition of free heads is mirror lookup** (Sec. 3.1 gives the corrected reading of the AdditivOrgan).
3. **True composition is delivered exclusively by ordinally structured organs.** To repair the triangle you do not change the lattice — you change the head family. The constraint is the organ design, and it must be ordinally *hard*; additivity alone is none (Proposition 1).

The ordinal recipe. How H_{ord} organs are built — three ingredients, each individually necessary (proven by ablation on subtraction): (1) a **bottleneck in the task dimension** — a nonlinear per-token encoder presses each operand to a scalar $v(t) = \phi(e_t)$, so the head sees only the task algebra of scalars (alone: held-out 0.000); (2) a **realizing, learnable init** at the parabolic form $r_c = 0.5c$, $\beta_c = -0.25c^2$ (frozen it jams at 0.36; absent, the optimizer never finds the line); (3) a **class-balanced loss** — rare answers are CE-cheap and tip systematically (with balance: held-out 1.000). The learned number line that emerges is an *attractor*: near-identical across seeds, monotone, quasi-equidistant ($-14.4 \dots 17.8$). Two independently designed architectures converged on this recipe (Sec. 3.4), and the harder structure wins on both size and generalization: 8.3k parameters, held-out 1.000. The recipe’s final step comes after training: **crystallization** (Sec. 3.9) — extract the learned invariant, install the exactly realizing form in its place; capacity, value range, and dose become deployment parameters, and the extraction’s fit quality doubles as a label-free authenticity verifier.

The form family extends across task classes without new machinery. For selection, max/min becomes a smooth function of the *same* learned number line — the algebraic identity $\max(x, y) = \frac{1}{2}(x + y + |x - y|)$ cast as organ structure:

$$u = \frac{1}{2}(v_a + v_b) + \frac{1}{2}\sigma|v_a - v_b|, \quad \sigma = \tanh(w_{\text{op}}^T P e_{\text{op}}), \quad \delta_c = s(cu - qc^2). \quad (2)$$

with the operator entering only as a sign and cross-operator transfer built in structurally. **One form family — the parabola over a learned number line — now carries add, sub, max, and min.** The principle behind the family is representation-theoretic, and Round 5 proves it constructively: for the cyclic relation $(a+b) \bmod 3$, the *same* learned number line is wound onto the circle —

$$\delta_c = s \cos\left(\frac{2\pi}{3}(u - c)\right), \quad u = v^T P e_a + v^T P e_b. \quad (3)$$

— a von-Mises profile in place of the parabola, no per-class freedom — and scores train 1.000, held-out 1.000 (Sec. 3.2). The structural form must carry the group of the task: $\mathbb{R}$ and the parabola for ordinal tasks, $\mathbb{Z}_3$ and the circle for cyclic ones. For median, the identity $\text{median} = \text{sum} - \text{max} - \text{min}$ composes the same max/min machinery over one number line.

The strict organ recipe. Everything above closes into one design, measured end to end in Sec. 3.2:

The strict organ recipe

Structural form	carries the group of the task: parabola/ $\mathbb{R}$, circle/ $\mathbb{Z}_n$, or multiplicative recurrence;
Map	pulls the task onto its canonical coordinate system: identity, logarithmic, or circular map;
Route gate	controls jurisdiction only, parameter-free by embedding or parser match;
Crystal	replaces the learned approximation by the exactly realizing form.

The result is an organ with **no free computing channel**: every generalization provably comes from structure. The crystal’s lattice is part of the law: snap to the lattice of the task’s group — $\mathbb{Z}$ for ordinal tasks, the log lattice for multiplicative ones, the $\mathbb{Z}_n$ circle for cyclic ones — verified per group by linear R^2 , log R^2 , or circular resultant (Sec. 3.10). The route-gate test makes that operational — free heads fall exactly onto their bare calibration ceiling (0.548 vs 0.55, digit for digit) while the structural form loses nothing, at bit-identical language NLL. MLP gates are a covert computing backdoor — a one-parameter path in ranking space that rescues only the 3-class case (Sec. 3.2, Round 7) — so strict organs take route gates, and every generalization claim in this regime must survive the route gate. Verification (invariant R^2), robustness, and deployment (support table, capacity, dose) are calculations thereafter.

2.3 The label-free harvest protocol

Harvesting a capability from the donor is a protocol, and every rule in it is a measured correction:

1. **Probe the donor on the collection metric, before collecting.** Free generation lies: the donor scores 0.000 on word-addition in free generation (it answers in digits) while its candidate-restricted logit ordering is *perfect* — accuracy 1.000, $p(\text{argmax})$ 0.97. The binding pre-collection check is a candidate-restricted mini-probe (24 prompts, ~ 75 s); free generation remains only a coarse format filter (it correctly caught a 0-shot yes-bias of $p \approx 0.79$ that made a yes/no answer space unusable).
2. **Design single-token candidate spaces.** Multi-token answer words (the BPE “teen” words) collide at the first token; the addw10 space (candidates *four. . . ten*) is collision-free by construction.
3. **Sharpen the donor distribution at $\tau=2$ before distilling.** The donor puts $p \approx 0.97$ on the right answer, but the diffuse residual mass dissolves hard structures: raw donor-KL holds only 0.25–0.67 on the hard form, $\tau=2$ (equivalently $T=0.5$ — found independently on both architectures, hence universal) restores ground-truth-quality training signal *without using ground truth*, reaching 1.000.
4. **Take class balance from the donor argmax** — also label-free.

The full chain uses zero ground-truth labels at every stage.

2.4 Gates and routing: the system protects itself

Gate taxonomy (measured). Memorizing organs fail by falling back to the host prior; the *difference gate* — answer released only where the organ moves the host away from its prior, i.e. the placebo contrast run at inference — releases 30/30 correct (forced error $0.306 \rightarrow 0.000$) and rejects 90% of unseen pairs. Compositional organs fail by ± 1 neighbors *while confidently firing*; they need the *margin gate* ($p_1 - p_2$ on the candidate slice). The compositional organ is **self-calibrated**: pooled over 3 seeds $\times$ 49 pairs, all three errors carry margins 0.02–0.04 against a correct-answer median of 1.00, so every threshold from 0.1 to 0.7 yields **error 0.000 at coverage 0.98** (forced error 0.020).

Routing. Organs without a jurisdiction gate annihilate each other: naively summing two organs collapses both tasks to chance (0.073/0.010). The task identity, however, is *linearly readable from the host’s own state*: a single linear probe on the layer-0 scan state — **384 parameters, three classes including “no organ responsible”** — routes at **1.000** on 96 held-out inputs. Routing costs no host training and no extra forwards, and abstention is a built-in class.

3 Results

All results on the living host (o1-state A1, 1,713,673 params, 909M streamed C4 tokens, frozen). All runs placebo-controlled (shuffled-target organ, same architecture and budget); dose-0 regression and unmount verified exact; language NLL reported per run. Training runs in 8–105 s on CPU.

3.1 Memorization, mirror lookup, and composition

Organ	Family	Params	Train	Held-out	Placebo (train/held)	NLL
Velocity MLP	free (state route)	66k	0.744	0.100	—	carry intact

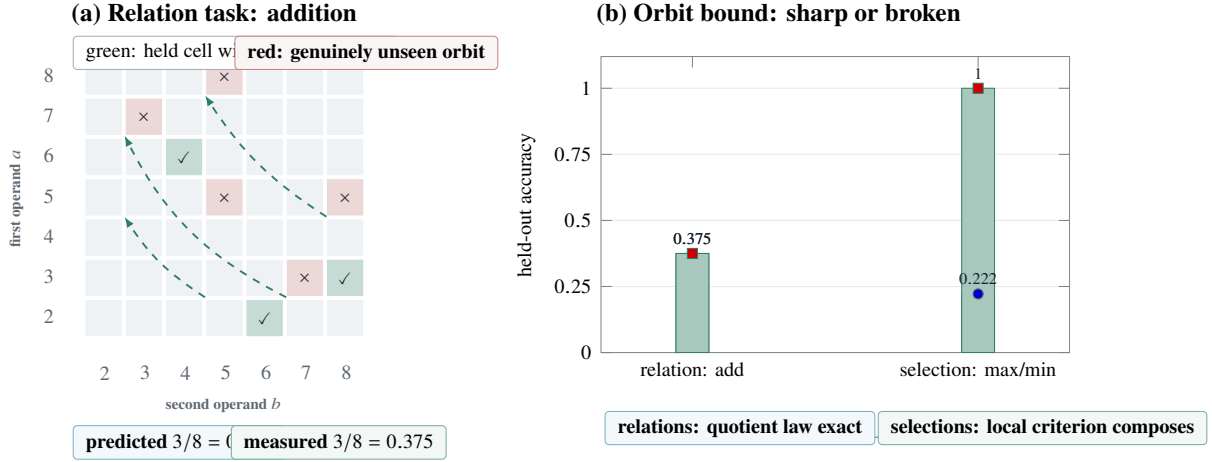


Figure 3. The Quotient Law and its exact scope. Left: the prospective addition test names every held cell before training. A symmetric free head answers exactly the cells whose $\mathbb{Z}_2$ orbit intersects the training lattice; all genuinely unseen orbits fail, identically across seeds. Right: the same orbit quantity is a guaranteed lower bound. It is sharp for relation tasks and structurally broken by selection tasks whose criterion factorizes class-locally.

Organ	Family	Params	Train	Held-out	Placebo (train/held)	NLL
AdditivOrgan, GT	H_{sym} (free symmetric)	8,686	1.000	0.900 (3/3 seeds)	0.128 / 0.100	8.6656 bit-identical
SubOrgan encoder form, GT	H_{ord}	12,498	0.968	1.000 (3/3 seeds)	0.161 / 0.000	8.6656 bit-identical
SubOrgan hard form, GT	H_{ord}	8,324	1.000	1.000 (3/3 seeds)	0.136 / 0.333	$\leq +0.0009$

The table stratifies exactly as the Quotient Law demands. The velocity MLP memorizes (the frequency gradient is its fingerprint, Sec. 4.1). The AdditivOrgan’s held-out 0.900 is **quotient generalization, not composition**: 9 of its 10 held-out cells have their mirror (b,a) in training — the identical input for a symmetric head — and the single truly unseen cell, 6+6, fails in 3/3 seeds, always by ± 1 ; a synthetic head without the host reproduces the same 0.900 including $6+6 \rightarrow -1$. What that experiment really demonstrated is symmetry identification plus memorization — and the correction sharpens the rest of the table: **the ordinal forms are the only organs that ever composed**. Held-out 1.000 on truly unseen orbits, at the smallest parameter counts in the table, including the 6+6 cell no free head can reach (the hard form closes it: add held-out 1.000 > 0.900).

3.2 The Quotient Law, confirmed prospectively and parameter-free

The law’s decisive test was registered before the run. Square lattice 2..8, 41 training cells, a mixed held-out set of 3 mirrored and 5 truly unseen cells (two both-sided pairs, {7,3} and {8,5}, plus the diagonal cell 5+5). The prediction file names the fate of every cell in advance, from pure orbit counting — no fitted parameter anywhere.

	Predicted (before measurement)	Seed 7	Seed 8
H_{sym} held-out accuracy	0.375 (= 3/8: exactly the mirrored cells)	0.375	0.375
Per-cell fate	6+4, 2+6, 3+8 correct; 7+3, 3+7, 5+5, 8+5, 5+8 wrong	16/16 as predicted	16/16 as predicted
H_{sep} null control	0.000 (trivial group — mirrors save nothing)	0.000	0.000

The measurement hits the prediction to the third decimal, per cell, both seeds identical. The error signature refines Proposition 3 on the spot: three of the five unseen failures are +1 — including the diagonal 5+5, the same signature as 6+6 in Sec. 3.1 — while the two {8,5} cells escape with +3 to the candidate boundary (*sixteen*), the class with only one-sided margins. The anchor negatives complete the picture: neither the diagonal nor a full anchor row $a=9$ lifts a free head above held-out 0.000. After orbit counting, no free-head result in this program survives as composition — and every ordinal result does.

Round 4 then tested the law’s scope on a third capability class — and both sides of the dichotomy were registered in advance. Selection (max/min via *larger/smaller*, 98 cells, both operators); held-out set of 9 cells spanning all four orbit types (mirrored, cross-operator, fully unseen, diagonal). The orbit bound for free heads is $2/9 = 0.222$; the registered predictions said all three head families *break* it (sym ≥ 0.778 , sep 1.000, hard 1.000), because on a selection task the answer *is* an operand and the local pair-membership solution is the SGD attractor (Proposition 4). Measured: **1.000 for all three families, all 9 cells, both seeds, 6/6 runs** — the bound broken by a factor of 4.5, on the same host infrastructure and measurement convention under which addition obeyed it to the third decimal. The two-point contrast is the prospective proof of the dichotomy:

	Round 3 (relation: addition)	Round 4 (selection: max/min)
Orbit bound (free heads)	0.375	0.222
Measured (free heads)	0.375 — bound sharp, per cell	1.000 — bound broken 4.5×

Relation tasks force structure into the head; selection tasks hand it to the optimizer for free. And the ordinal hard form scores 1.000 on both sides of the table.

Round 5 makes it three task families per side. On the relation side, modular addition — *a remainder b is r*, $r = (a+b) \bmod 3$, 42 training cells, 7 held-out cells spanning all orbit types (`w6_r5_mod3.json`): the cyclic hard form — the learned number line wound onto the circle — hits its registered prediction exactly at **train 1.000, held-out 1.000** (2/2 seeds, 7/7 cells). No other family even fits the training lattice: symmetric, separable, and the ordinal parabola all top out at train ≤ 0.93 and land at held-out 0.857 with the *identical* single error — the mirrored cell 6+3, both seeds, every family. The diagnosis chain behind that number produced two discoveries at once (`w6_gitter_gesetz.md` Sec. 10). First, the program’s exactest confirmation of orbit identification: the training misfit forms the cluster $\{3,6\} \times \{3,6\} \rightarrow \text{two}$, and the held-out cell (6,3) inherits precisely the (3,6) training error — **the orbit transfers the trained state, not the truth**. Second, the rescue of the truly unseen cells was never the free head: bare heads calibrate at train 0.55, held-out 0.000 on this task (deterministic across seeds and embedding widths), and switching the organ’s 4-gram gate off collapses held-out to 0.000. **The gate is a computing channel** (law v4.2): an organ’s “free family” is never bare — it is $\delta \times \text{gate} + \text{host}$, and $\text{argmax}[h_c + g(a,b) \cdot \delta_c]$ is non-separable in the pair, strictly richer than the free head alone. Its reach is set by class granularity: it rescues relation tasks with a coarse answer space (mod-3, 3 classes: 0.857) and none with a fine one (addition, 13 classes: Round 3 sat exactly on its orbit bound 0.375). Class granularity is the dichotomy’s third axis, and the design consequence is immediate: a strict organ needs a gate that only routes, never computes. On the selection side, median of three — *a b c middle is ...*, 197 training cells, 13 held-out cells including two complete six-cell permutation orbits, orbit bound $1/13 \approx 0.077$ (`w6_r5_median.json`): the free heads answer **all 13 at 1.000** — sym and sep, every seed, exactly as registered — the bound broken thirteen-fold. Median is full selection, no intermediate class: the multiset code $x_a + x_b + x_c$ plus per-class linear separation carries the middle, and the hard form built from the max/min identities over one number line reaches held-out 1.000 wherever its optimization converges (the $|\cdot|$ -kink jam is an optimization property, registered in advance and reported separately from structure). The dichotomy now stands on three task families per side: add, sub, and mod-3 are relations; max, min, and median are selections — and on every one of the six, the family that carries the task’s structure sits at 1.000.

Round 6 locates the boundary inside selection (law v4.3). Two operator tasks, predictions committed in advance. *Closer-to* — *a b near t is*, the operand nearer the anchor, 258 training cells, 12 held-out cells including a deliberate factor hole: operand 6 never co-occurs with anchor 3 in training, six cells deleted (`w6_r6_closer.json`). Measured: the separable head and the hard form (a soft-argmin over one number

line) score 1.000/1.000 in every seed, and the factor hole is bridged in all six organ runs, 9/9 in bare-head calibration before them — the anchor tilt is learned as smooth structure, not as a table; the original hole thesis fell in calibration and the committed anti-thesis hit. The round’s single deviation — sym2 seed 8 escaping to the anchor on two cells — is a newly catalogued failure mode, not a bound. *Parity-switch* — *a either b is*: min if $a+b$ is even, else max — a criterion that depends on the co-player and therefore does not factorize per class (`w6_r6_parity.json`). Measured: the symmetric head trains at 1.000 and keeps membership perfect — the answer is one of the operands at rate 1.0 in every seed — while its *choice* on unseen cells falls to the membership floor (0.5 and 0.0 across seeds; the random-switch control sits exactly on the committed floor 0.667); the separable head loses even mirrored cells for the first time on a selection task; a closed-form symmetric solution exists — features $(p_c, c \cdot p_c)$ — and is not the SGD attractor; and the committed no-rescue prediction for the gate held, because a scalar gate is class-uniform and cannot reorder the δ ranking. Only the π -product hard form — $u = \frac{1}{2}(v_a + v_b) - \pi/2 \cdot |v_a - v_b|$ with $\pi = \tanh(w \cdot e_a) \cdot \tanh(w \cdot e_b)$ — scores **1.000/1.000**, and on the random-switch control it correctly jams in training: there is nothing to carry. **Law v4.3**: selection decomposes into *membership*, which generalizes always — at rate 1.0 even under a random criterion — and *choice among members*, which generalizes freely exactly when the criterion factorizes class-locally (max, min, median, closer-to); pair-coupled criteria re-install relation behavior inside the choice component. The decisive task property is the factorizability of the criterion, not the answer set.

Round 7 measures the granularity axis — and it is a step, not a slope. Round 5 left the gate rescue’s reach open between 3 classes (rescued) and 13 (not). The registered prediction said the rescue rate on truly unseen cells decays smoothly with block occupancy $49/m^2$, with the step alternative named as the explicit discriminator — the step won and the prediction fell (`w6_r7_granular.json`, with the mod-4 follow-up in `w6_r7_mod4.json` locating the edge): U-rescue **4/4 at mod-3, then 0/4 at mod-4, mod-5, and mod-7 alike**, with held-out falling exactly onto the orbit inheritance at every $m \geq 4$ (0.286 — precisely the mirrored cells; on mod-7 even those inherit the training misfit) and addition sitting on its bound at 13. The gate rescue exists only in the coarsest class space and is completely dead from four classes upward. The mechanism replaces the occupancy model: the MLP gate is a **one-parameter path in ranking space** — an interpolation between host prior and organ delta — and the set of argmax classes reachable along a single path is small: at 3 classes the right class almost always lies on the path, from 4 it combinatorially does not. The positive control scales where the trick cannot: the cyclic hard form scores **1.000/1.000 on $\mathbb{Z}_5$ and $\mathbb{Z}_7$** (4/4 runs) — the group representation is scale-proof.

The route gate finishes the separation: structure computes, route gates. The gate is made parameter-free — a hard embedding match on the operator token ($\cos > 0.9$), no gradient, no computing channel (`w6_r7_routegate.json`). Both registered predictions hit. The cyclic hard form loses *nothing*: train 1.000, held-out 1.000, gate activation 0.000 on text and 1.000 at the read point, host NLL 8.6656 bit-identical. The free head falls honestly to held-out 0.000, its train accuracy landing at 0.50/0.55 — **digit for digit the bare calibration ceiling** measured without host or gate. The hidden computing channel of v4.2 is thus cleanly detachable at zero cost to structure, and the design closes into the strict organ recipe (Sec. 2.2): MLP gates are a covert computing backdoor; strict organs take route gates; and every generalization claim in this regime must survive the route gate.

3.3 The route comparison: logit shell dominates every column

Same donor targets, same dose, two routes:

Route	Eval	Held-out
Velocity (state-space)	0.531	0.333
OrganShell (logit space)	0.969	0.750

Both seeds numerically identical per route. The velocity route remains a tool for state dosing; as a capability route it is retired by this table.

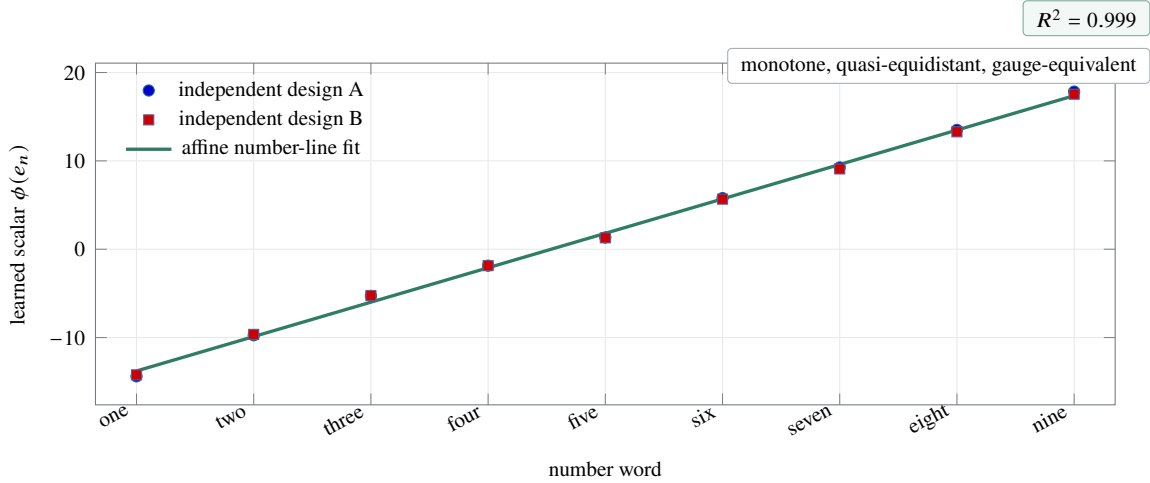


Figure 4. The number line is an attractor. Two independently designed dual organs, trained from separate seeds, converge to the same one-dimensional ordering. Their scale and offset are gauge freedoms; the task-bearing invariant is the affine ordinal geometry.

3.4 Two independent designs converge — twice

Subtraction was solved twice, by two agents working independently from the same failure data:

- **Encoder form** (12,498 params): 1D bottleneck + realizing init + class balance → held-out 1.000, 3/3 seeds; the learned number line is near-identical across all three seeds (the attractor, visible directly in the artifact).
- **Hard form** (8,324 params): structural ordinality $\delta_c = \text{scale} \cdot (c \cdot u - q \cdot c^2)$ → held-out 1.000, 3/3 seeds, independently replicated.

Both then built the **dual organ** — one organ, both operations, operator switch — as a deliberate convergence test. Both passed at ceiling:

Design	Params	add train/held	sub train/held	Seeds	NLL
Dual hard form (GT)	8,357	1.0 / 1.0	1.0 / 1.0	3/3	$\leq +0.01$
Dual encoder form (GT)	14,625	1.0 / 1.0	1.0 / 1.0	2/2	8.6656 bit-identical

The hard form learns the operator sign from the token itself: $\text{sgn}(\text{plus}) = +0.987$, $\text{sgn}(\text{less}) = -0.999$. Seed 8 learns switch *and* number line consistently inverted — genuine gauge freedom of the form (Sec. 4.4). A causal ablation settles the canonical question: **one number line carries both operations** (the frozen subtraction line already gives addition 0.917 train; joint fine-tuning closes the residual anisotropy).

3.5 The label-free chain, end to end

Candidate-restricted donor targets (352 prompts, donor accuracy 1.000, p 0.97) distilled by candidate-KL into the organ of the living host, **zero ground-truth labels anywhere**:

Stage	Result
Donor-KL → AdditivOrgan	eval 0.969 (seeds 7 and 8 numerically identical), placebo 0.240/0.146, held-out 0.750, NLL 8.6656 bit-identical, 34 s CPU
Donor-KL, $\tau=2$ + argmax balance → dual organ	train 1.000 , held-out 1.000 (2/2 seeds), NLL 8.6656 bit-identical

The AdditivOrgan row is itself the Quotient Law to the letter: exactly its three mirrored held-out cells are correct, and the one truly unseen cell — the diagonal 3+3 — fails by -1 in both seeds. The label-free

composition claim therefore rests where it must: on the ordinal dual organ, which answers its held-out cells at 1.000.

The $\tau=2$ sharpening does not merely match ground-truth training — on this space it *replaces* it at equal quality (0.969 vs 1.000 GT-train on the 16-pair space; 1.000/1.000 on the dual organ).

3.6 The organ as a procedure: train on length 2, deploy unbounded

Everything above transplants *functions* — fixed-arity maps from operands to an answer. The **chain organ** transplants a *procedure*. It is a recurrent accumulator with its own $O(1)$ scalar state: an operator token flips a sign, an operand token accumulates its number-line value $v(t)$, and the hard-form readout fires at the *is* token. 5,732 parameters.

It is trained **exclusively on chains with two operators** (“three plus two less one is”) and evaluated on chain lengths it has never seen (96 tasks per length):

Chain length	2 (trained)	2 (held-out structures)	3	4	6	8	12
Recurrent organ (5,732 params)	1.000	1.000	1.000	1.000	1.000	1.000	1.000
Flat control (5,187 params)	1.000	1.000	0.042	—	—	—	—

The control isolates the mechanism: a flat slot-form organ with the *same* hard-form readout and matched capacity is perfect at the training length and falls to exact chance (0.042) one step beyond it. Recurrence — not capacity, not the readout form — carries length generalization. The prediction was registered before the run (recurrent $L3/L4 \geq 0.8$; flat = chance) and was exceeded at every length.

This is the host’s foundation F6 — *train short, deploy unbounded* (Foss, 2026a) — reproduced at organ level: the organ inherits the $O(1)$ -state principle of the organism it serves, and with it the first **algorithmic** capability in the organ stack, unbounded multi-step arithmetic from bounded training.

And the procedure itself is harvested label-free. The same organ distilled purely from the donor (candidate-KL, $\tau=2$ sharpening, argmax balance — zero labels; the donor’s own accuracy on the chains is 0.961, chains are not trivial even for the 27B) from 256 length-2 chain rows extrapolates across every length: **L2 0.938, L3 0.948, L4 0.917, L6 0.969, L8 0.885, L12 0.833** (96 fresh chains each; ground-truth reference 1.000 throughout). The donor supplied nothing but one-step distributions over the end values of length-2 chains; at no point in the chain — harvest, training, architecture — does a label or a long chain appear. The organ inherits exactly the donor’s residual uncertainty, and the length generalization comes entirely from the organ’s structure — and Sec. 3.9 then removes that inheritance completely: crystallized, the same organ answers every length at 1.000.

A second procedure confirms the rung. A counting organ (~600-parameter relevant core), trained on sequences of 2–4 content words (“cat dog bird is three”), scores **1.000 on lengths 5–7, 8–10, and 11–12** (3×48 fresh sequences from 20 content words) — and its learned increment $g(e_i)$ is constant at **3.065 ± 0.001 across all 20 content words**: content-invariant cardinality, the procedure in its purest form, with the constant itself gauge-free. Counter and operator-controlled accumulator are the first two rungs of an organ-level Chomsky ladder — Sec. 3.7 climbs the third, Sec. 3.8 the fourth and fifth. Both procedures are H_{ord} organs; under the Quotient Law they sit with the hard and encoder forms among the only organs that compose.

3.7 The stack organ: rung three, and capacity as a deployment parameter

The ladder’s third rung is true LIFO, placed against the lineage of differentiable external-memory and stack-augmented recurrent systems. The task: a sequence of *open* ⟨symbol⟩ and *close* events; at the read point, the question is which symbol lies on top of the stack (*open cat open dog close open bird is* → *bird*). A $k=1$ scalar cannot carry this — the state must hold an unbounded suffix of the history — so the stack organ gets a slotted state $s \in \mathbb{R}^{k \cdot 16}$ with shift structure:

$$\text{push}(x) : s \leftarrow A_{\text{push}}s + BPe_x, \quad \text{close} : s \leftarrow A_{\text{pop}}s, \quad \text{score}(y) = (Pe_y)^T Ws. \quad (4)$$

Following the canon recipe, A_{push} and A_{pop} are initialized at the realizing form — block-shift matrices — and remain learnable; the control is the same organ with freely initialized A matrices (12,288 parameters each). Training never exceeds depth 2.

The registered prediction fell, and its fall was the discovery. The prediction said free matrices memorize (depth 3 near chance 0.125). Measured: free hits 0.979/0.990 on depths 3/4 (`w9_stack_organ.json`), and still 1.000 under the cleanest regime, max-depth ≤ 2 throughout training (`w9_stack_maxtiefe.json`). The mechanism attack resolved it within the day: the end-depth generator makes the answer mostly the *last-pushed* symbol, so a recency tracker with no LIFO at all passes that metric. This is the third measurement hook the program has caught this way (after the H-01 probe and the addw candidate space), and it is why the two-instrument protocol exists: a prediction registered in advance flags the anomaly the moment it falls, and an adversarial metric settles what the naive one could not ask.

The buried-symbol test asks the real question: push k symbols, then pop $k-1$ — the answer is the *first* symbol, buried under the entire history, and no recency signal survives. Alongside it runs the algebraic discriminator $\text{pop} \circ \text{push} = \text{id}$. Both separate cleanly (`w9_stack_mechanismus.json`): the shift-structured organ scores **1.000** at its full depth with $\text{pop} \circ \text{push}$ functional error 0.141 ($\approx$ identity); the free matrices score **0.083** — chance is 0.125 — with $\text{pop} \circ \text{push}$ error 0.504. The free organ never built a stack. This fixes the scope of Proposition 4 in its final form: the selection character of top-of-stack makes the *task* free-learnable, never the *algorithm* — wherever a flat shortcut exists, SGD takes it, and the structural prior is what buys the true LIFO. For functions, the ordinal prior was the difference between lookup and composition; for procedures with memory, it is not optional at all.

The Capacity Law. Where does the depth limit sit? The registered prediction: a step function exactly at the slot count of the state geometry. Measured at two capacities (`w9_kapazitaets_gesetz.json`), training unchanged at max-depth ≤ 2 :

Buried depth	3	4	5	6	7	8
$k = 4$ slots (12,288 params)	1.000	1.000	0.135	0.188	—	—
$k = 6$ slots (23,552 params)	1.000	1.000	1.000	1.000	0.188	0.146

Slots 3 through 6 were never loaded during training — depth never exceeded 2 — and they carry at 1.000. Capacity comes from state geometry, not from data: push and pop are block permutations, hence depth-equivariant, and the learned parts P , W , B operate only at the top slot. Every depth is the top slot of its own future, so training at depth 2 trains every depth the geometry can hold.

Capacity surgery. The law’s consequence was tested the same day, prediction registered in advance (buried-8 ≥ 0.9): take P , W , B from the 4-slot training, embed them in an 8-slot geometry, and set $A_{\text{push}}/A_{\text{pop}}$ to the exact block shifts of the new size. Measured (`w9_kapazitaets_chirurgie.json`): **buried 4, 6, and 8 all at 1.000; buried 9 at 0.125 — exactly chance, exactly at the new slot boundary — with zero gradient steps.** Capability capacity is a deployment parameter. An organ is trained once, at depth 2, and shipped at whatever capacity the deployment demands; the OrganBank (Sec. 3.14) can inflate a cold organ to target capacity at load time, at the cost of writing two permutation matrices. The invariant geometry is dimensioned at shipping, not at training — the stack’s version of train-short-deploy-unbounded, with the bound itself now a free parameter.

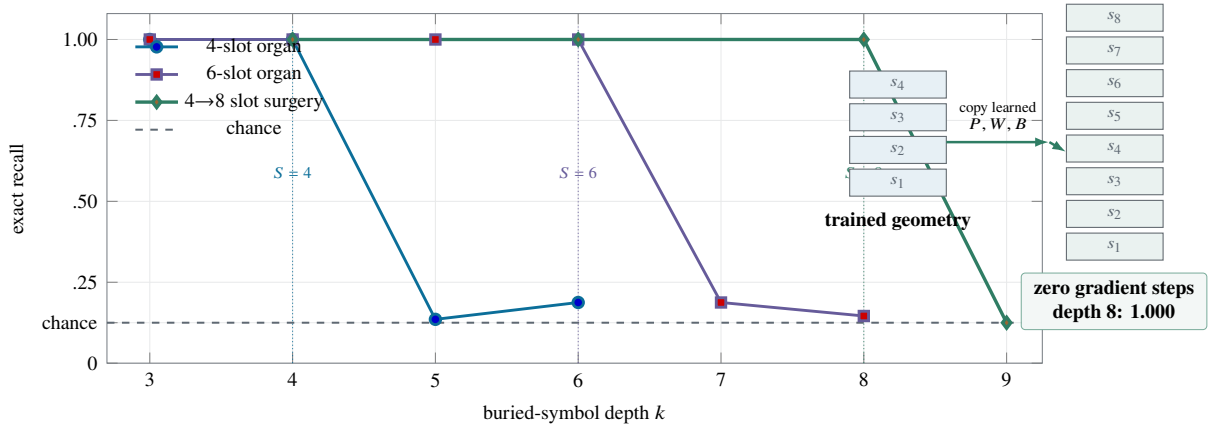


Figure 5. Capacity is a deployment parameter. Buried-symbol recall is a step function at the number of shift slots, including slots never activated during depth ≤ 2 training. Transplanting the learned parameters of a 4-slot organ into an exact 8-slot shift geometry moves the edge to depth 8 with zero retraining.

3.8 Rungs four and five: two parameters per rung

The ladder’s fourth rung is the canonical context-sensitive language $a^n b^n c^n$ — and the invariant principle prices it exactly, in advance: no stack, but two $O(1)$ counters, $c_1 = \#a - \#b$ and $c_2 = \#b - \#c$, with balanced $\iff c_1 = c_2 = 0$ and the hard-form readout

$$\text{score}_{\text{yes}} - \text{score}_{\text{no}} = s_0(\theta - c_1^2 - c_2^2). \quad (5)$$

The organ has **two learnable parameters** — s_0 and θ . Everything else is the invariant. Trained only on $n \leq 3$ (plus unbalanced perturbations), with the registered prediction $n = 4.8 \geq 0.95$ including the near-balanced cases, it measures (w11_kontextsensitiv.json): **train 1.000; $n = 4$ through 8 at 1.000; near-balanced confusers — exactly one symbol off by one — 1.000; c_2 -confusers 1.000.** The structural control is a one-counter organ that sees only c_1 : on the constructed $a^n b^m c^n$ confusers — $\#a - \#b = 0$ but $\#b - \#c \neq 0$, exactly the cases only the second counter can see — it was predicted to fall to chance and falls past it to **0.000**, deterministically blind, because the confusers are built to satisfy its only counter. The blindness is structural, not statistical.

Rung five crosses the dependencies. The copy language ww — is the second half the first half again? — is the canonical mildly context-sensitive language, the TAG class of linguistics, and it defeats both existing mechanisms by construction: a stack reverses order (LIFO decides ww^R , not ww), and counters are order-blind. The invariant principle delivers the Rabin–Karp fingerprint as an organ — the ladder’s first *multiplicative* recurrence, $h \leftarrow r \cdot h + v(\text{sym})$, the separator switching registers, same $\iff h_1 = h_2$ — again with **two learnable parameters**. Trained only on halves of length ≤ 4 (w13_kopier_organ.json): train 1.000, mixed never-seen lengths 6–10 at 1.000, and **permutation confusers at 1.000** — same multiset, different order, the exact case no counter can see. The counter control proves the mechanism: perfect on flips, exactly **0.000** on permutations — it answers *same* on every one, as registered.

The fingerprint’s capacity is an arithmetic table, not a measurement campaign. The multiplicative recurrence doubles the leading term per step, so float32 must lose flip positions where 2^{L-p} falls below $\text{ulp}(2^L)$. Registered prediction: a positional edge at 23 ± 1 , independent of length. Measured (w13_praezisions_kante.json): every position up to 24 detected at 1.000, position 25 at 0.41 — **a sharp positional edge at 25, length-independent; float64 detects all 30 positions at 1.000.** The prediction missed by one — the mantissa carries 24 bits including the implicit one, not 23 — while the mechanism is confirmed exactly. The deployment rule reads like the stack’s: the organ’s capacity stands in the arithmetic table (float32 ~ 24 positions, float64 52), and the fix is register width or a double register (r and $1/r$). Across the ladder, robustness is now uniformly a calculation: verification from the invariant R^2 (Sec. 3.9), capacity from the state geometry (Sec. 3.7), arithmetic precision from the mantissa table.

The ladder climbs by recurrence class — additive $k=1$ (counting, operator-controlled chains), shift matrices (LIFO), additive $k=2$ (context-sensitive), multiplicative $k=2$ (mildly context-sensitive) — and its

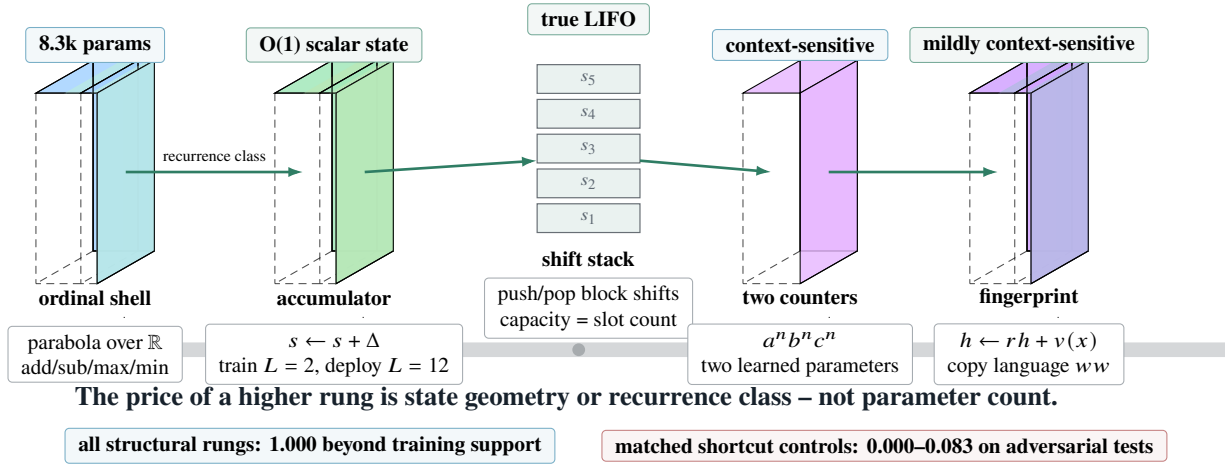


Figure 6. The capability ladder. The program climbs from finite ordinal forms to an additive accumulator, an exact shift stack, two independent counters, and a multiplicative fingerprint recurrence. Each rung adds the minimal state geometry required by its language class; the top two rungs use only two learnable parameters.

economics are the paper’s cleanest: **the price of a Chomsky rung is a state dimension or a recurrence class, never parameters.** Two learnable scalars decide a context-sensitive language, and two more decide the copy language. Every rung inherits train-short-deploy-unbounded from the host’s F6, and every rung’s state geometry is subject to capacity surgery (Sec. 3.7): the ladder is dimensioned at deployment. And the classes compose — Sec. 3.11 nests the multiplicative parser inside the additive accumulator and breaks the vocabulary’s number limit with it.

The ladder has outgrown its teacher. Before building the rung we probed the donor on it, twice, under the standard pre-collection protocol (candidate-restricted yes/no on $a^n b^n c^n$ balance in the same word format; artifacts `donor_chunk_ccs_sonde.npz` / `donor_chunk_ccs2_sonde.npz` on the harvest host, logged as entry 8.112 of the program record). At $n = 2..5$ with 4 shots the 27B donor answers *yes* on 24 of 24 prompts (accuracy 0.583 — exactly its yes-rate); at $n = 1..3$ with 6 shots, *yes* on 24 of 24 again (accuracy 0.500 — again exactly the yes-rate). Under both format levers the donor is a pure bias null model on this task: there is nothing to harvest, because one cannot distill what the teacher does not know. This is the converse of Sec. 3.9’s donor-space result — there the transplanted organ exceeded the donor’s *space*; here the organ ladder exceeds the donor itself: **two learnable scalars decide, with unbounded length extrapolation, what 27 billion parameters cannot answer in the same format.** Capability sources are therefore two — harvest where the donor can, structural construction where it cannot — and the bank’s intake control verifies both the same way, by their invariant.

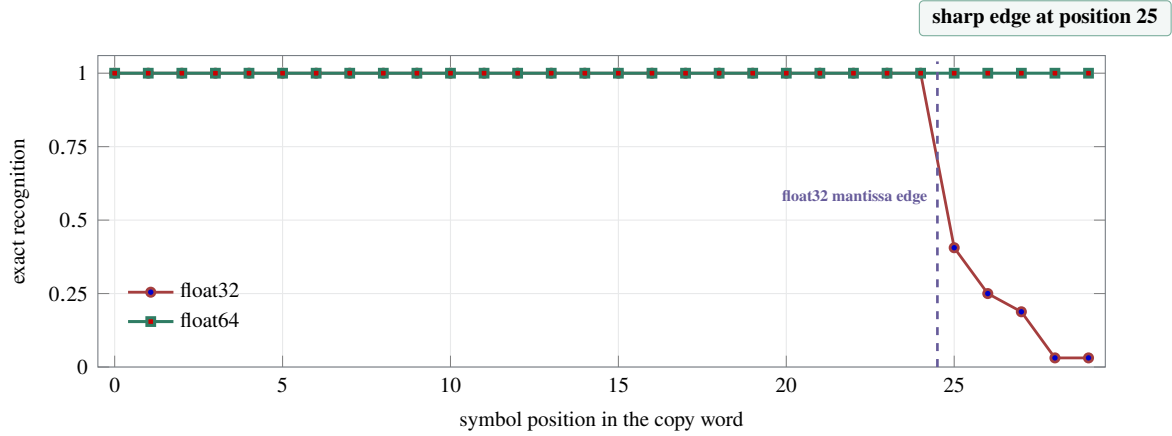


Figure 7. Numeric precision is the exact capacity law of the fingerprint organ. Float32 remains perfect through position 24 and collapses sharply at 25, while the identical recurrence in float64 stays exact throughout the measured range. Capacity is set by representation precision, not training length or parameter count.

3.9 Crystallization: train the invariant, install the exact form

Capacity surgery (Sec. 3.7) generalizes into the program’s second great lever. After training, the learned readout *approximation* is replaced by the exactly realizing construction over the learned *invariant* — train, extract the invariant, install the exact form. The extraction is label-free (a least-squares self-calibration over the training support), and afterwards capacity (slots), value range (candidates), and dose (S0) are all deployment parameters. We call the step **crystallization**, and it enters the recipe as its fifth ingredient. Three measurements establish it.

The negative fixes the mechanism first. Extending the dual organ’s answer range by least-squares extrapolation of its *fine-tuned readout rows* fails, and fails diagnostically (`w3_skalen_chirurgie.json`): the trained rows no longer lie on the hard-form manifold (weight-fit R^2 0.804 — the hard form was the init, not a constraint), so the extrapolated rows `c=11..16` tip to the candidate boundary (new sums: 0.091, every error an escape to *sixteen*) and drag the existing range down to 0.59. The mirror image of Sec. 3.7: surgery succeeds exactly when the structure is exact; learned structural approximations do not extrapolate.

So transplant the invariant, not the rows. The content-bearing learned object is the number line ϕ , and it is linear to R^2 0.998. From $\phi(\text{one}..\text{nine})$ alone, α and γ are fitted by least squares — zero labels, zero donor calls — and *every* readout row `c=4..16` is constructed exactly realizing, with dose `S0=10` chosen because the hard-form argmax is scale-invariant while the host prior is not (dose sweep in `w3_skalen_chirurgie_dosis.json`: `0.5` $\rightarrow$ `0.44`, `10` $\rightarrow$ `1.000`). Result (`w3_skalen_chirurgie_v2.json`): **all 67 addition sums answered at 1.000 end-to-end — including all 33 sums 11–16, which neither the organ nor the 27B donor has ever seen** — with the host NLL bit-identical at 8.6656. The registered prediction (new sums raw ≥ 0.85) is exceeded to the ceiling.

Integer crystallization makes the donor distillation lossless. The donor-distilled chain organ of Sec. 3.6 carries the donor’s residual uncertainty (baseline down to 0.844 at length 12, `w7_kette_kristall.json`; the donor’s own chain accuracy is 0.938). Diagnosis reads the invariant directly: over the true operand tokens *one..nine* the line fits at R^2 0.998, and the single outlier is $v(\text{ten}) = 0.90$ — a token that never occurs as chain input, so ϕ never shaped it; the crystallization map is fitted over the training support. Affine crystallization lifts L2–L6 to 1.000 (L12 0.969) and extends the candidate space to 1..16 for free. The invariant, however, is *discrete* — so the map snaps to the integer lattice, $v = \text{round}((\phi - \gamma)/\alpha)$, with the registered prediction *all 1.000*. Measured: **L2 through L12, every length 1.000, on the extended candidate space** — from a teacher that itself scores 0.938. Rounding acts as digital error correction over analog learned values: the curvature residual that accumulates along long chains is removed structurally, not statistically. The pipeline noisy 27B distribution $\rightarrow$ organ $\rightarrow$ crystal now ends in an exact algorithm.

The fit is a verifier. If crystallization measures whether the organ truly learned the structure, then its fit quality authenticates a capability without asking a single task question. Registered prediction: placebo $R^2 < 0.7$. Measured (`w7_kristall_verifier.json`): the genuine donor-distilled organ reads out at

R^2 **0.99848**; three placebo organs — identical training on row-permuted donor distributions — read out at **0.321, 0.216, 0.005**. Verification moves from a task sample to a structural property: the OrganBank (Sec. 3.14) runs this as its intake rule — crystallize every incoming organ, store the exact form when the invariant is clean, mark a memorizer as such *before it ever answers*. The stack has the same instrument already measured in Sec. 3.7: the pop ◦ push functional error, 0.141 versus 0.504, is this verifier in algebraic form.

The map family closes: multiplication is addition under log. If structural forms are group representations, the multiplicative group $(\mathbb{R}_+, \cdot)$ demands no new organ class — only a different map in the same $O(1)$ accumulator, with the readout on the log lattice: $\delta_c = s_0((\log c)a - q(\log c)^2)$. Trained on nothing but two-factor products (w10_mul_organ.json): the learned v fits $\alpha \cdot \log n + \gamma$ at R^2 **0.999** against 0.927 for the linear fit; the registered linear-lattice falsifier breaks on three-factor chains exactly as predicted (0.021); and integer crystallization on the log lattice delivers **length-3 and length-4 factor chains — never seen in training — at 1.000**. The raw pre-crystal organ carries chain curvature (L3 0.687, below its registered ≥ 0.9 — the one clause of the prediction that fell), and crystallization removes it: the same lossless step, on the third group. The map family stands: one accumulator structure, maps as group isomorphisms — linear for $(\mathbb{R}, +)$, log for $(\mathbb{R}_+, \cdot)$, the circle for $\mathbb{Z}_n$ — the group-representation principle extended from functions to procedures, constructively. One learned number line — wound onto the circle, kinked through $|\cdot|$, or mapped through log as the task’s group demands — now carries add, sub, max, min, median, closer-to, parity-switch, mod-3, and mul.

And the transplanted capability exceeds its donor’s space. The full label-free chain runs on multiplication end to end (w10_donor_mul.json): probe 12/12 under the pre-collection protocol, harvest of two-factor products ≤ 10 only (single-token candidates *one..ten*, 184+23 prompts, donor accuracy 1.000), donor-KL at $\tau=2$ with argmax balance, then log crystallization. Raw: L2 1.000, L3 0.509, L4 0.035 — the invariant underneath: $r_{\log}^2$ **0.99887** — crystallized, on candidates 1..16: **L2, L3, and L4 all 1.000**, every clause of the registered prediction hit. The donor supplied nothing but two-factor distributions over products up to ten; the crystallized organ computes four-factor chains up to sixteen. This is the first transplantation that exceeds the donor’s space on both axes at once — structure (two factors $\rightarrow$ four) and value range (ten $\rightarrow$ sixteen) — with zero labels. The distillation transfers the invariant, not answers: **the donor is a map teacher, not an answer teacher**. And for multiplication, crystallization is not polish — the raw approximation decays to 0.035 at length 4 while the invariant sits at 0.999 — it is what makes the capability reliable at all. Probe $\rightarrow$ harvest $\rightarrow$ KL $\rightarrow$ crystal is now the standard path for algorithmic capabilities.

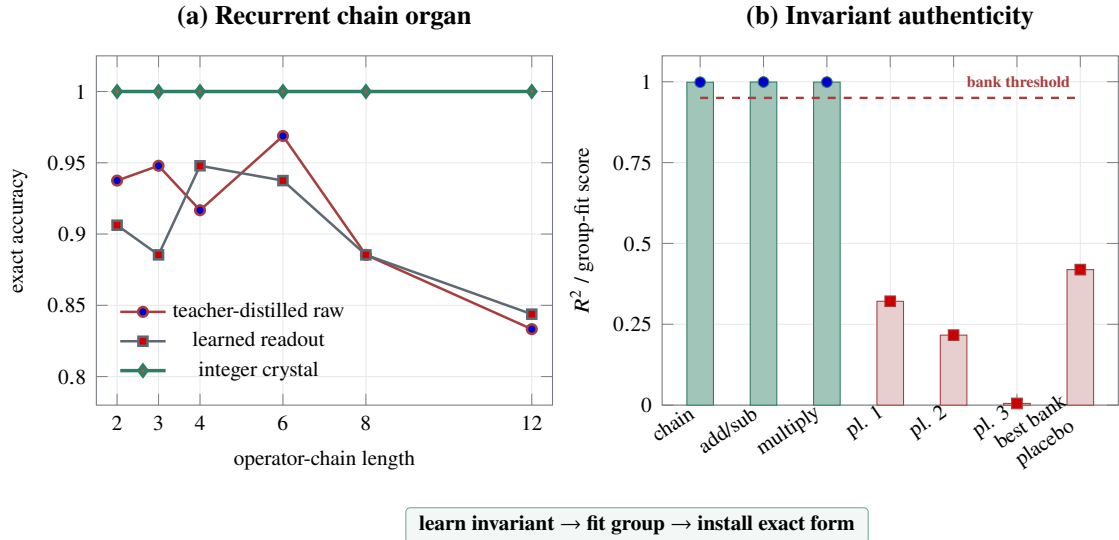


Figure 8. Crystallization makes label-free distillation lossless. Left: the learned and donor-distilled chain organs carry residual approximation error; replacing the readout by the exact integer-lattice realization yields 1.000 accuracy from length 2 through 12 without labels or gradient steps. Right: the invariant fit is also a task-free authenticity test. Genuine additive, multiplicative, and chain organs cluster at $R^2 \approx 0.999$; placebo memorizers remain far below the bank threshold.

3.10 The signal threshold and the format bridge

The map-teacher thesis has a validity boundary, and we went looking for it. For the bank’s third organ — mod-3 in word form — the donor probes came back weak: 0.458 at 4 shots, 0.417 at 8; not a bias null model (chance 0.33), but far below harvest standard. Instead of aborting, the registered hard test staked the thesis at its riskiest point: the cyclic hard form has almost no degrees of freedom, donor noise that does not fit the circle is unrepresentable, so distillation can only extract the number line or fail — prediction: invariant $R^2 \geq 0.95$ despite the weak teacher. **The prediction fell** (w12_mod_organ.json): donor accuracy on the training rows 0.389, invariant R^2 0.323, crystal 0.370. The mechanism is clean: the structural constraint filters noise that does not fit the form, but it cannot extract a map where the signal carries no consistent number line — **a 0.39-teacher teaches no map**. Against the 0.938-teacher anchor (invariant R^2 0.998, crystal 1.000, Sec. 3.9), the map-teacher thesis gains its validity boundary: distillation is lossless above a signal threshold — between 0.39 and 0.94, exact location open — and dead below it.

The threshold is a property of the format, not of the donor — and the bridge crosses it. The same 27B that scores 0.389 on mod-3 in word form scores 0.965 in digits. The format bridge harvests where the teacher is strongest and builds the organ where the host lives: class-aggregated targets — P_k as the mean digit-space answer distribution over all $N \equiv k$ — read from the *existing* digit harvest (352 rows, class purity 3/3), distilled into the word organ with **zero new donor calls**. Measured (w12_bruecke.json): **all 81 cells at 1.000, 27 of them held out**. The 0.39-teacher was dead; the same donor over the bridge is perfect. The class distribution is format-free — only the prompt was weak — so the harvest protocol gains its final rule: harvest in the teacher’s strongest format.

The crystal lattice must be the lattice of the task’s group. The registered integer snap then destroyed the perfect organ — 1.000 → 0.35, linear invariant R^2 0.077 — and the diagnosis found not damage but geometry: v is a *circle* map, with circular class centers 0.52 / 1.51 / 2.51, spacings 0.994 / 1.001 / 1.005, resultant 0.9998 (w12_kreis_kristall.json). The circular phase snap crystallizes the same organ at **81 of 81 = 1.000**. The clause enters the recipe: snap to $\mathbb{Z}$ for ordinal tasks, to the log lattice for multiplicative ones, to the $\mathbb{Z}_n$ circle for cyclic ones — with the invariant verifier per group: linear R^2 , log R^2 , circular resultant. The bank’s map recognition (Sec. 3.14) is complete on all three measured groups.

The bridge has a second mode, and it revives dead capabilities. Table knowledge needs no map: the per-value bridge reads each answer distribution from the digit harvest directly. Primality — a capability that had died in word form — comes back at **13 of 13 correct, zero labels, zero gradient steps**, with

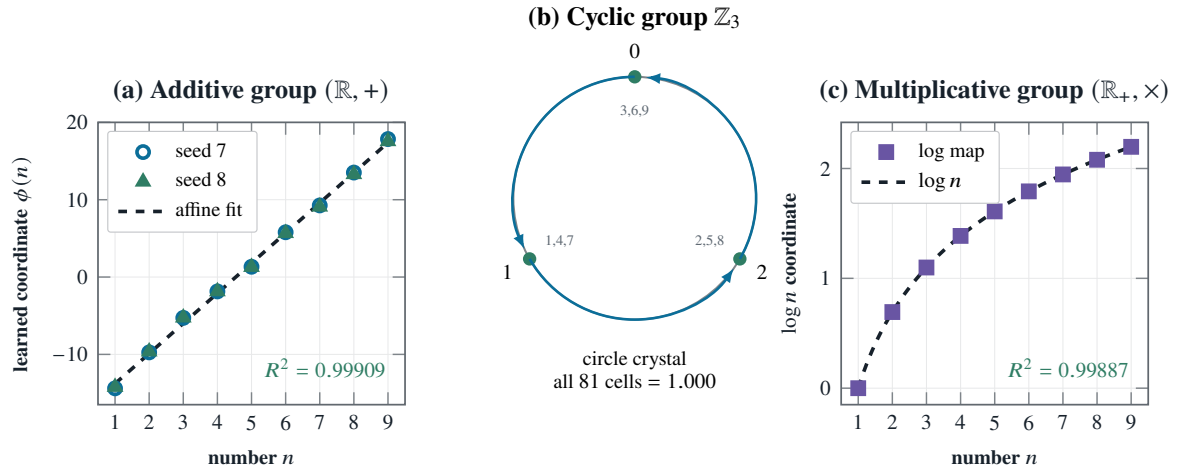


Figure 9. One learned coordinate supports multiple task groups. Additive and subtractive tasks use an affine number line, modular tasks wrap that coordinate onto a circle, and multiplicative tasks use a logarithmic lattice. The OrganBank identifies the carried algebra by the best invariant-map fit.

structural abstention on the three values never harvested (`w12_prime_bruecke.json`). The companion experiment writes a bank rule: lending the arithmetic dual’s number line to the prime table works exactly on that invariant’s support 1..9 and reads deterministically wrong outside it — every error at $n \geq 10$. **Shared invariants need a support register:** each invariant declares its validity interval, and the intake control checks coverage before lending.

3.11 Unbounded input, exact computation, unbounded output

The sixteen-limit falls structurally. The host’s number vocabulary ends at *sixteen*, and every task so far lived inside it. The decimal parser breaks the input limit by composing two recurrence classes of the ladder, nested: a multiplicative parser $h \leftarrow 10 \cdot h + v(\text{digit word})$ — the fingerprint class of Sec. 3.8 — running *inside* the additive accumulator of Sec. 3.6, with the operator token closing the segment ($\text{acc} \leftarrow \text{acc} + \text{sgn} \cdot h$, $h \leftarrow 0$). *one two plus three is* parses as $h = 12$, $\text{acc} = 12$, $h = 3$, $\text{acc} = 15$. Measured (`w14_dezimal_parser.json`): parser train 1.000 and unseen two-digit patterns 1.000 *raw*; digit invariant over zero..nine at R^2 0.99866; the crystallized parser reads *one two* = 12, *two one* = 21, *nine zero* = 90, and **one zero zero = 100** exactly — arbitrary magnitudes from a ten-word digit vocabulary. The order-blind control is structurally dead: 0.297 train, 0.000 on the place-swap confusers, as registered. One registered sub-clause fell, and its mechanism is already a law of this paper: snapping v to the integers while keeping the *learned* readout drops that mixed evaluation to 0.2 — the same scale mismatch as the fine-tuned-rows negative of Sec. 3.9. Crystallization is atomic, map and readout together, never half; the exact parser probe above and the end-to-end result below are the full-crystal proof. And the ladder gains its composition clause: **recurrence classes nest, and crystallizing the inner invariant carries the whole composition.**

The output limit falls from the structure side. Multi-token generation was the one output mode the program had never conquered — every organ so far answered in a single token, because free generation against the host prior loses. It falls not by better sampling but by dosing. The chain: the `w14` parser — the *only* learned component in the entire chain, one digit line at R^2 0.999 — reads the operands; exact multiplication runs on crystallized integers; divmod decomposes the result into digits; and each digit is emitted token-wise as a shell dose onto the digit-word slice of the living host. Registered prediction: 1.000 at dose 10. Measured (`w15_mehrstellig.json`): **96 of 96 random products ($a \in 10..99$, $b \in 2..9$, results up to 891 — far beyond any vocabulary number word) sequence-exact at 1.0000, zero errors.** The organism generates multi-digit answers correctly because the organ doses the output stream step by step instead of hoping. Two-by-two-digit multiplication is the same mechanism by construction: both operands through the parser, one more emission step.

The arithmetic line of the organism is complete: **unbounded input, exact computation, unbounded**

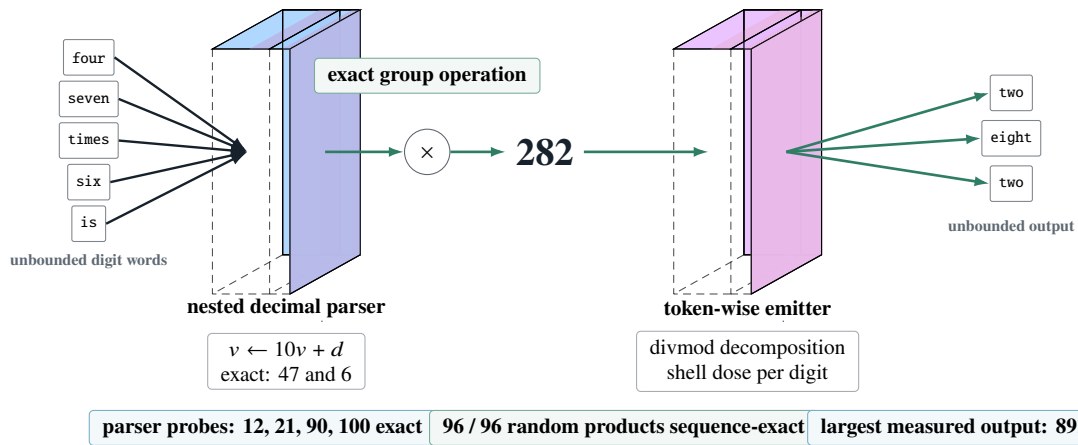


Figure 10. Vocabulary limits are not capability limits. Two nested affine recurrences parse arbitrarily long digit-word sequences into exact integers; the crystallized group operation computes on those integers; token-wise dosed emission decomposes the result back into a sequence. The living host generates all 96 measured multi-digit products exactly.

output — from ten digit words, one learned line, and three exact forms.

3.12 Gates and router

Component	Result
Margin gate (compositional organ)	error 0.000 at coverage 0.98 , stable across thresholds 0.1–0.7 (147 pooled judgments; forced error 0.020)
Difference gate (memorizing organ)	30/30 released correct; 90% of unseen pairs rejected; free error 0.103 vs forced 0.388
State router (384 params, 3 classes incl. abstention)	1.000 on 96 held-out

One refinement from the prospective lattice test feeds directly into gate design: errors on truly unseen cells can escape to the candidate *boundary* (the +3 flights to *sixteen* in Sec. 3.2), and the boundary class has only one-sided margins — it evades the interior margin calculus, so boundary classes receive their own margin treatment.

3.13 The system: Ship v4 to Ship v6

`wesen_demo_v4.py` composes the entire chain — living host + dual donor-KL organ (0 labels) + state router + margin gate (threshold set data-driven at 0.01 from the margin audit; minimum correct margin 0.016):

Full task space, 16 additions + 35 subtractions: 51 released, 51 correct (100%), 0 abstentions needed. 30/30 text windows routed to the host. Host NLL 8.6656 — the shell is mount-free; the host is never touched. Runtime 4.8 s on CPU.

The artifact chain is persisted and verified: `organ_dual_donor.pt` (digest `e6931667f93eff05`, 14,625 params, deterministically reproducible from seed 7 in 28 s), a loading API that verifies the digest, passes an 8/8 held-out self-test, and abstains on the router negative test (*times* → no answer).

Ship v5 replaces the single mounted organ with the bank. The system is now: living host + OrganBank holding **two cold, digest-verified, crystallized 27B organs** — the arithmetic dual on $(\mathbb{R}, +)$ (`e6931667f93eff05`) and the multiplication organ on $(\mathbb{R}_+, \cdot)$ (`a8e293ed7675ba63`), both admitted through the group-classifying intake control of Sec. 3.14 — + a 3-class state router (387 parameters). Measured (`wesen_demo_v5.json`):

90 of 90 tasks correct — 16 additions, 35 subtractions, 33 two-factor and 20 three-factor multiplication chains — with the route correct on 90 of 90, including the three-factor prompts whose 6-token length the router has never seen. 30/30 text windows to the host. Host NLL 8.6656 bit-identical. Runtime 4.9 s on CPU.

The system’s defining sentence at this stage: *two times two times four is sixteen* has not a single training instance anywhere in the chain — the donor saw only two-factor products up to ten, the host has never seen a digit, and no label was used at any stage.

Ship v6 is the complete arithmetic of the organism in one stream. Four organs from three groups: the arithmetic dual on $(\mathbb{R}, +)$, the multiplication organ on $(\mathbb{R}_+, \cdot)$, the mod-circle organ on $\mathbb{Z}_3$ — built over the format bridge and the circle crystal of Sec. 3.10 — and the decimal line of Sec. 3.11 for multi-digit input and generated multi-digit output. Measured (wesen_demo_v6.json):

152 of 152 tasks correct — 16 additions, 12 subtractions, 33 multiplication chains, 67 mod cells, and 24 multi-digit products whose digit-word answers the organism generates, up to three digits. Route correct on 152 of 152; 30/30 text windows to the host; host NLL 8.6656 bit-identical. Runtime 8.0 s on CPU.

The demo line of the night: *four seven times six is* → *two eight two*. The build carries a method lesson, documented over two iterations: the state router is recency-dominated and does not separate format variants of the same task (137/152, then 135/152). The fix follows the strict recipe, not a bigger router: **the router routes coarsely** — task versus text, at 1.000 since v4 — **and the parser dispatches exactly**, operands and operators read deterministically from the parse, the answer format following from the input format. Minimize guessing; let structure decide. The ship line reads **v3 16/16 (one organ) → v4 51/51 (dual organ + gate) → v5 90/90 (group-classified bank) → v6 152/152 (complete arithmetic, three groups, generated output)**.

3.14 The OrganBank: capabilities live cold

Organs do not need to be resident. The OrganBank stores them as digest-addressed cold artifacts on disk — 65 KB for the dual organ — behind a registry, the state router, and cold-load on demand. On a mixed 30-input stream: **20/20 tasks answered correctly, 10/10 text windows routed to the host; latency 1.1 ms route / 0.9 ms cold load (digest verification included) / 1.0 ms answer**. The cold load costs less than the router forward itself, so capability residency is free by construction: a bank of 1,000 organs is ~65 MB cold at ~3 ms end-to-end per access.

This is the host’s foundation F4 — the two-system law (Foss, 2026a) — realized on the capability axis: the organism keeps its knowledge in cold, digest-verified pages and now keeps its skills in cold, digest-verified organs — the same architecture on two axes.

The bank’s door is the crystallization verifier of Sec. 3.9, hardened into an admission rule (organbank_eingang.json): digest verification plus invariant $R^2 \geq 0.95$. The genuine dual organ enters at R^2 0.99844 and is registered together with its crystallized deployment parameters; a ϕ -permuted copy of the same organ — the no-training negative control, statistics preserved, invariant destroyed — is rejected as a memorizer at R^2 0.019.

And the door reads more than authenticity: it classifies the group. Fitting every map and letting the best R^2 win labels an incoming organ with its algebra (organbank_karten.json). The naive fit first rejected the genuine mul organ ($r_{\log}^2$ 0.931) — the same support trap as $v(\text{ten})$ in Sec. 3.9: *nine* never occurs as a mulw10 operand, and the bank cannot know an incoming organ’s training support — so the fit is trimmed leave-one-out. Hardened against three freshly trained placebo memorizers: **arith-dual admitted as $(\mathbb{R}, +)$ at linear R^2 0.99909, with the trimmed log fit staying below at 0.948; mul-log admitted as $(\mathbb{R}_+, \cdot)$ at 0.99887; the defect and all three placebos rejected at ≤ 0.42** — a margin of 0.53 between the most convincing memorizer and the threshold. The intake control is a group classifier: it determines, label-free, *which algebra* an incoming capability carries — and whether it carries one at all.

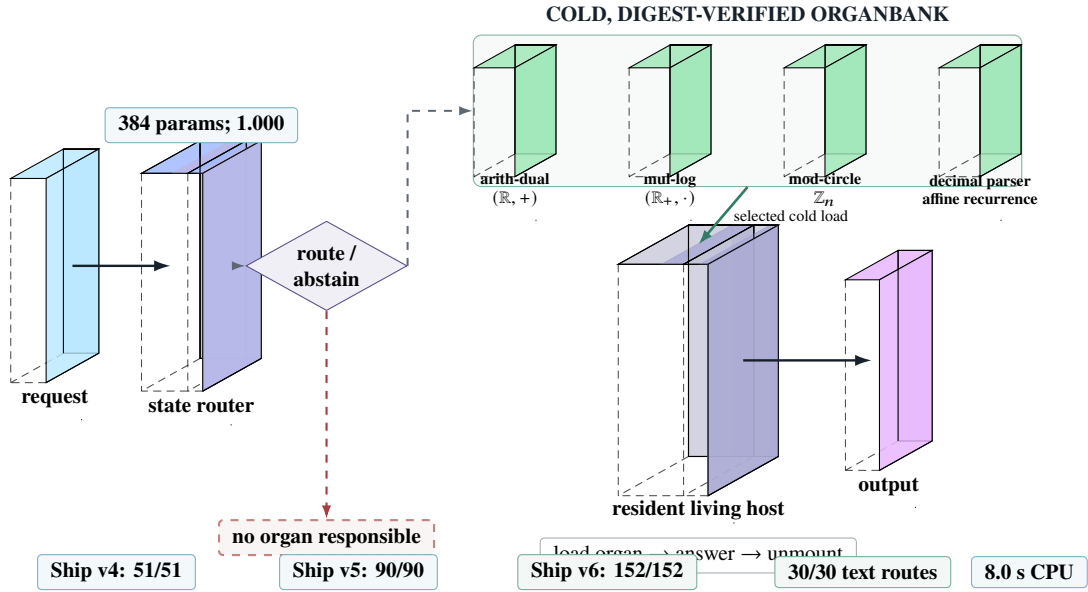


Figure 11. Capabilities live cold. The resident host and a 384-parameter state router stay in memory. Digest-addressed organs remain cold until the router and parser identify a jurisdiction; the selected shell is loaded, applied, and discarded. The final four-organ bank routes every measured arithmetic task and every text window correctly at unchanged NLL.

3.15 Deployment hardening: organs survive int4

QAD (Liquid AI, 2026) applied to the host (fake-quant int4 per-channel with STE, forward-KL against its own fp32 self, 400 steps, 22 s): NLL fp32 8.6656 $\rightarrow$ PTQ 8.6671 $\rightarrow$ **QAD 8.6654** (117% of the quantization gap closed). The donor organ on the QAD-int4 host: **16/16** — organs survive deployment quantization of their host unchanged, because the shell reads logits and embeddings, which stay fp32 under the standard layer recipe. Host int4 + fp32 organ shells is the fleet deployment form.

4 The Mechanism Story

Every negative result in this program was driven to its mechanism, and each mechanism became a clause of the law. This section is the causal chain.

4.1 Why every state-space route memorizes

The 66k velocity organ’s held-out failure (0.100) decomposed cleanly: any organ that receives the operand pair as a joint input has a memorizing solution available, and gradient descent takes it (the frequency gradient is the fingerprint: pairs seen $>6\times$ score 0.90). The route that structurally cannot see the pair — layer-0-only velocity — cannot learn at all, because the frozen head does not translate state pushes into token preferences. The resolution is not a better state route but a different space: the logit shell — where the Quotient Law then determines which constraint actually suffices. Not pair-blindness (Proposition 1 refutes it), but ordinal hardness.

4.2 The geometry detour, and the rule it produced

Subtraction first failed with a seductive geometric explanation: the frozen host’s number-word embedding *differences* are noise-dominated (mean coherence of equal-difference vectors: 0.15–0.32, versus 0.70 for the sum manifold). The explanation was measurable — and wrong as a limit. An exact closed-form solution existed all along (the number words are linearly independent in 128d; w with $w \cdot e_t = t$ fits to $2e-6$ and solves all 35 pairs). The failure was the *optimization path* (the linear form follows the noisy difference directions instead of the existing 1D solution) plus *class imbalance* (large differences are rare, hence CE-cheap; errors sit systematically one too low). This yielded a standing methodological rule: **prove exact-solvability before indicting geometry.**

4.3 Two falsifications, one law

The Quotient Law was not guessed — it was forced out by two registered falsifications, each attacked by protocol.

The trail opened with a control experiment: addition, the “solved” task, collapses identically on a triangular training lattice (train 1.000, held-out 0.000, 2 seeds). **Round 1** tested the pure drift theory against it — prediction: the diagonal’s self-loops pin the drift parity and repair the triangle. Measured: held-out 0.000, with errors up to ± 5 — falsified. The lesson is structural: memorization leaves the ordinal family entirely, so polytope bounds protect only ordinal solutions, never free heads. **Round 2** tested the linear capacity threshold — prediction: at embedding width 8 the permuted-target probe collapses below 0.7. Measured: the probe fits 0.957, width-independent — falsified. The correct accounting is the margin inequality system of Proposition 1, not dimension counting.

The synthetic mapping that followed — bare head families, no host, no gate — produced the law in one sweep: held-out accuracy equals the mirror share of the held set, exactly, on every lattice tried. **Round 3** then registered the law’s per-cell predictions before the measurement and was hit to the third decimal (Sec. 3.2). In the residue, the predecessor law’s variable — lattice skew — dissolved: the triangle fails not because it is *skewed* but because the mirror of every triangle cell lies outside the lattice — and the square never composed, it was mirrored. The operations were messengers; the orbits were the law.

4.4 The number line is an attractor with a gauge

Across seeds, architectures, and both operations, the learned representation converges to a monotone, quasi-equidistant number line — and the hard form exposes its symmetry: one seed learns line and operator switch jointly inverted, with identical behavior. The organ does not approximate arithmetic; it *finds the one-dimensional structure that is arithmetic*, up to gauge.

4.5 The scale anchor

The last donor-KL instability on the hard form was traced to the harvest data, not the architecture: 12 addition prompts (operands 2–5) are too weak a scale anchor for the *shared* number line against 205 subtraction rows — a λ -seesaw with scale compression above sum 8. KL on addition alone reaches 1.000; ground truth on the full space reaches 1.0/1.0 with no tuning. The harvest space, not the organ, sets the anchor — a rule now in the harvest protocol.

5 Related Work

Knowledge distillation. Knowledge distillation compresses a teacher into a student whose identity *is* the distillation product. It does not preserve a pre-existing host, does not operate label-free into a frozen recipient, and does not cross a 15,750 $\times$ parameter gap into a 1.7M-parameter organism whose language distribution must remain bit-identical.

Quantization-aware distillation. Quantization-aware distillation (Liquid AI, 2026) is self-distillation along the precision axis: the same model, BF16 $\rightarrow$ int4, capability presupposed. We use QAD exactly for deployment hardening of the host (Sec. 3.15). It is orthogonal to, and composable with, capability transplantation. The two axes meet in the measured deployment form: QAD-int4 host, fp32 organ shells.

Task arithmetic and model merging. Task arithmetic and model merging edit weights between same-architecture, same-size models. Donor and host here share no architecture, tokenizer, or dimension, and the host’s weights are never edited.

Adapters and LoRA. Adapters and LoRA inject trainable modules *inside* the host’s forward pass and train them on labels. Organs live outside the forward pass as a mount-free shell, train on a donor distribution with zero labels, carry their own abstention and routing machinery, and detach bit-exactly.

The problem solved here—label-free, placebo-controlled, reversible capability transplantation across four orders of magnitude into a living $O(1)$ -state host—is addressed by none of these lines. Where this system stands, it stands alone.

6 Reproducibility

Every number in this paper is a JSON artifact next to the script that produced it, in `mirkoNN-analysis/s3/`: `w3_komposition.json`, `w3_sub_encoder.json`, `w3_sub_hartform.json`, `w3_dual_hart.json`, `w3_dual_organ.json`, `w2_donor_additiv*.json`, `w4_abstinenz_gate.json`, `w5_zustand_s_router.json`, `w6_gitter_test.json`, `w6_gitter_test2.json`, `w6_gitter_test3.json`, `w6_klasse3_test.json` (each with the registered prediction inside the artifact), `w6_r5_mod3.json`, `w6_r5_median.json`, `w6_r6_closer.json`, `w6_r6_parity.json`, `w6_r7_granular.json`, `w6_r7_mod4.json`, `w6_r7_routegate.json`, `w7_ketten_organ.json`, `w7_laengen_extrem.json`, `w7_donor_kette.json`, `w7_kette_kristall.json`, `w7_kristall_verifier.json`, `w8_zaehl_organ.json`, `w9_stack_organ.json`, `w9_stack_maxtiefe.json`, `w9_stack_mechanismus.json`, `w9_kapazitaets_gesetz.json`, `w9_kapazitaets_chirurgie.json`, `w11_kontextsensitiv.json`, `w12_mod_organ.json`, `w12_bruecke.json`, `w12_kreis_kristall.json`, `w12_prime_bruecke.json`, `w13_kopier_organ.json`, `w13_praezisions_kante.json`, `w14_dezimal_parser.json`, `w15_mehrstellig.json`, `w3_skalen_chirurgie.json`, `w3_skalen_chirurgie_v2.json`, `w3_skalen_chirurgie_dosis.json`, `w10_mul_organ.json`, `w10_donor_mul.json`, `organbank_eingang.json`, `organbank_karten.json`, `organ_mul_donor_check.json`, `qad_wirt.json`, `wesen_demo_v4.json`, `wesen_demo_v5.json`, `wesen_demo_v6.json`, `wesen_organbank.json`; the full statement and proofs of the Quotient Law are in `s3/w6_gitter_gesetz.md`. The two donor probes on $a^n b^n c^n$ balance live on the harvest host as `donor_chunk_ccs_sonde.npz` and `donor_chunk_ccs2_sonde.npz`, logged as entry 8.112 of the program record. The shipped organ is a digest-verified 14,625-parameter file (`organ_dual_donor.pt`, sha256-16 e6931667f93eff05) reproducible deterministically from seed 7 in 28 seconds on CPU; the bank’s second organ, the donor-distilled mul-log organ, is persisted by `mul_organ_persist.py` with digest a8e293ed7675ba63 after passing the intake control (`organ_mul_donor_check.json`). The host checkpoint, its training life, and its theoretical guarantees are disclosed in the o1-state repository (Foss, 2026a).

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